

ISUZU

EMERGENCY RESPONSE GUIDE

2025 NRR EV



**400V
Li-ion**

 **WARNING:** Operating, servicing or maintaining this vehicle, can expose you to chemicals, including phthalates and lead, which are known to the State of California to cause cancer and birth defects or other reproductive harm. To minimize exposure, wear gloves or wash your hands frequently when servicing your vehicle.

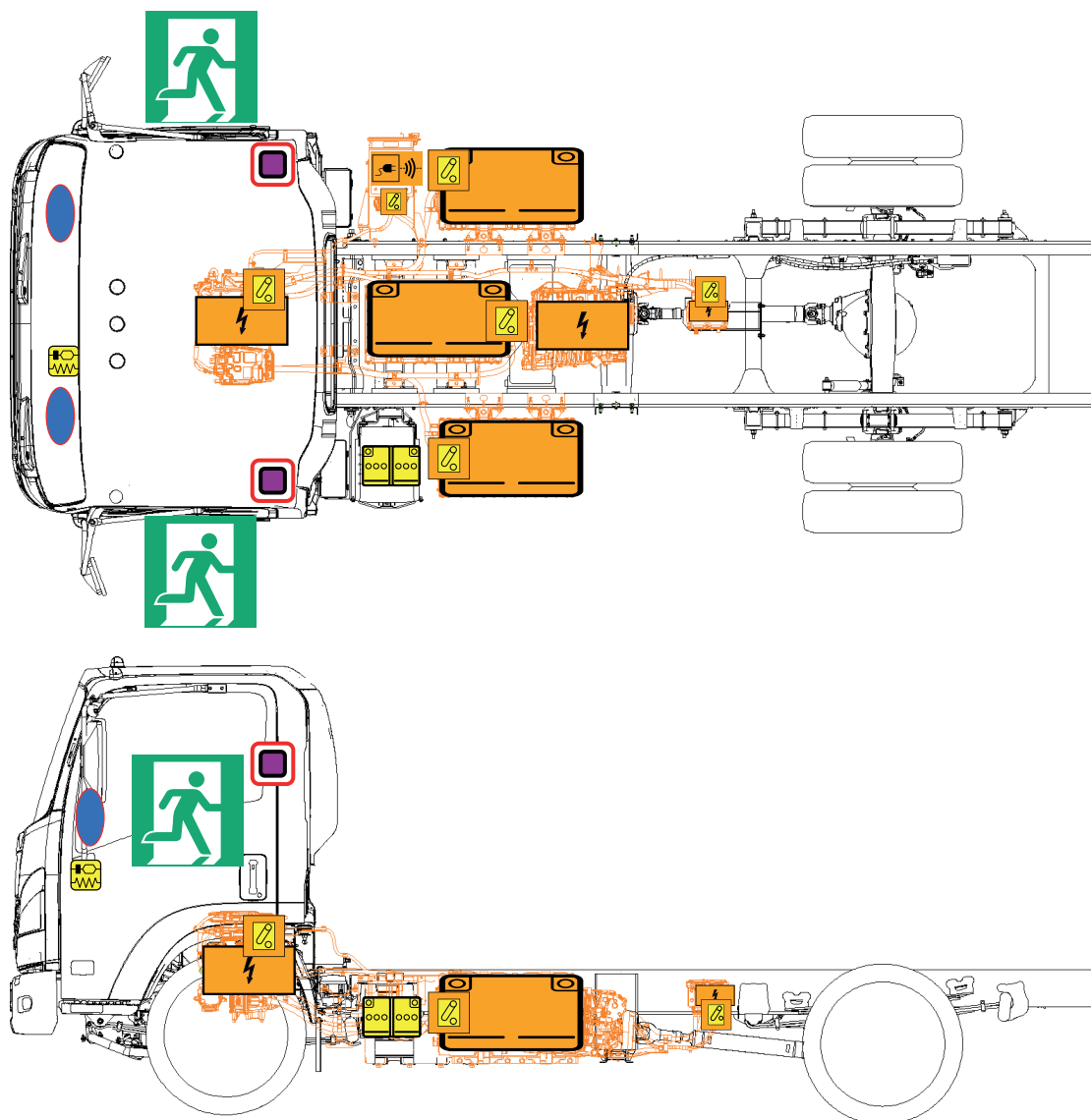
 **WARNING:** Battery posts, terminals and related accessories contain lead and lead compounds, chemicals known to the State of California to cause cancer and reproductive harm. Batteries also contain other chemicals known to the State of California to cause cancer. Wash your hands after handling.

FOR MORE INFORMATION ON THE WARNINGS VISIT
THIS WEBSITE:
www.P65Warnings.ca.gov

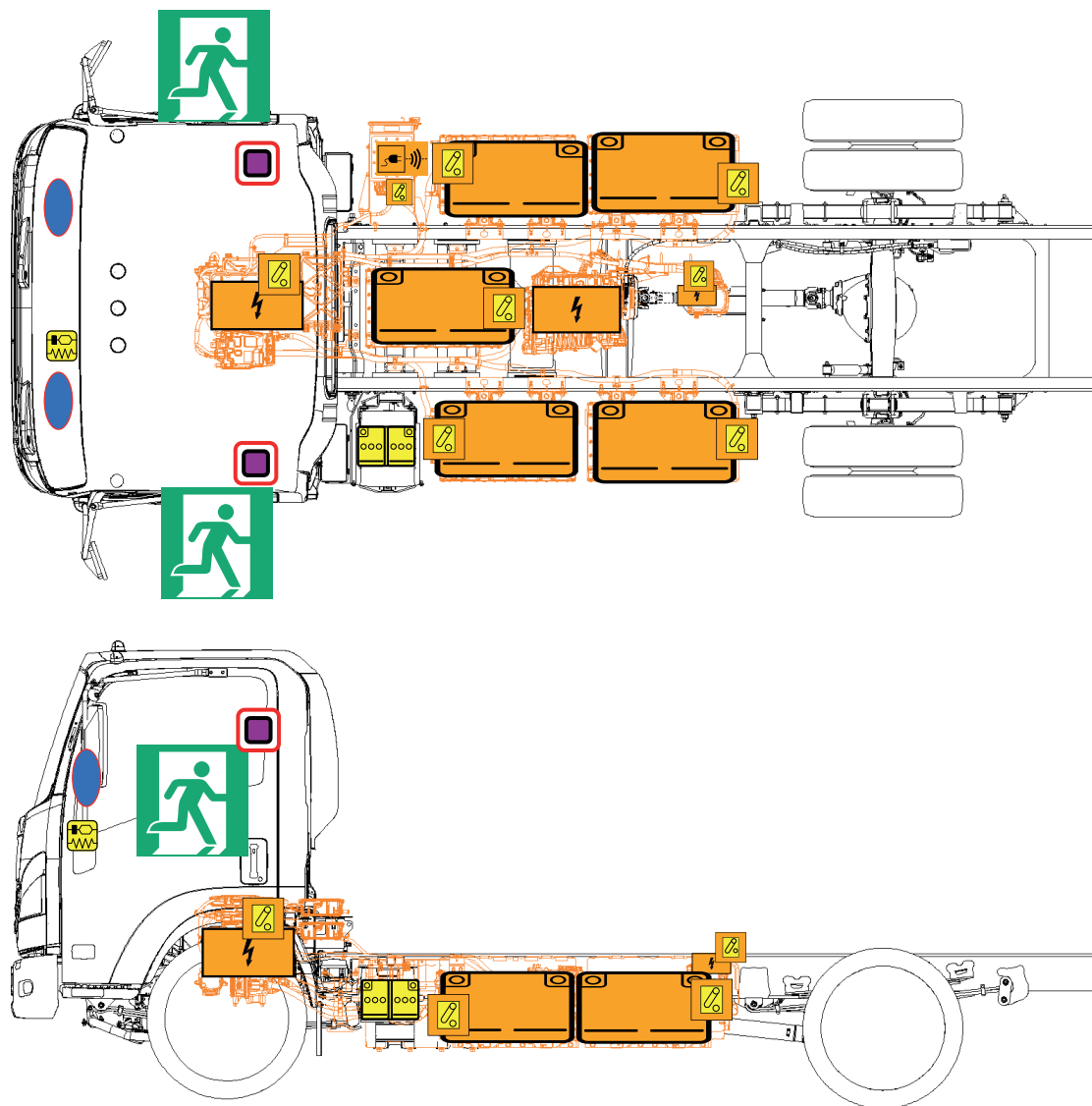
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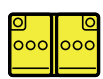
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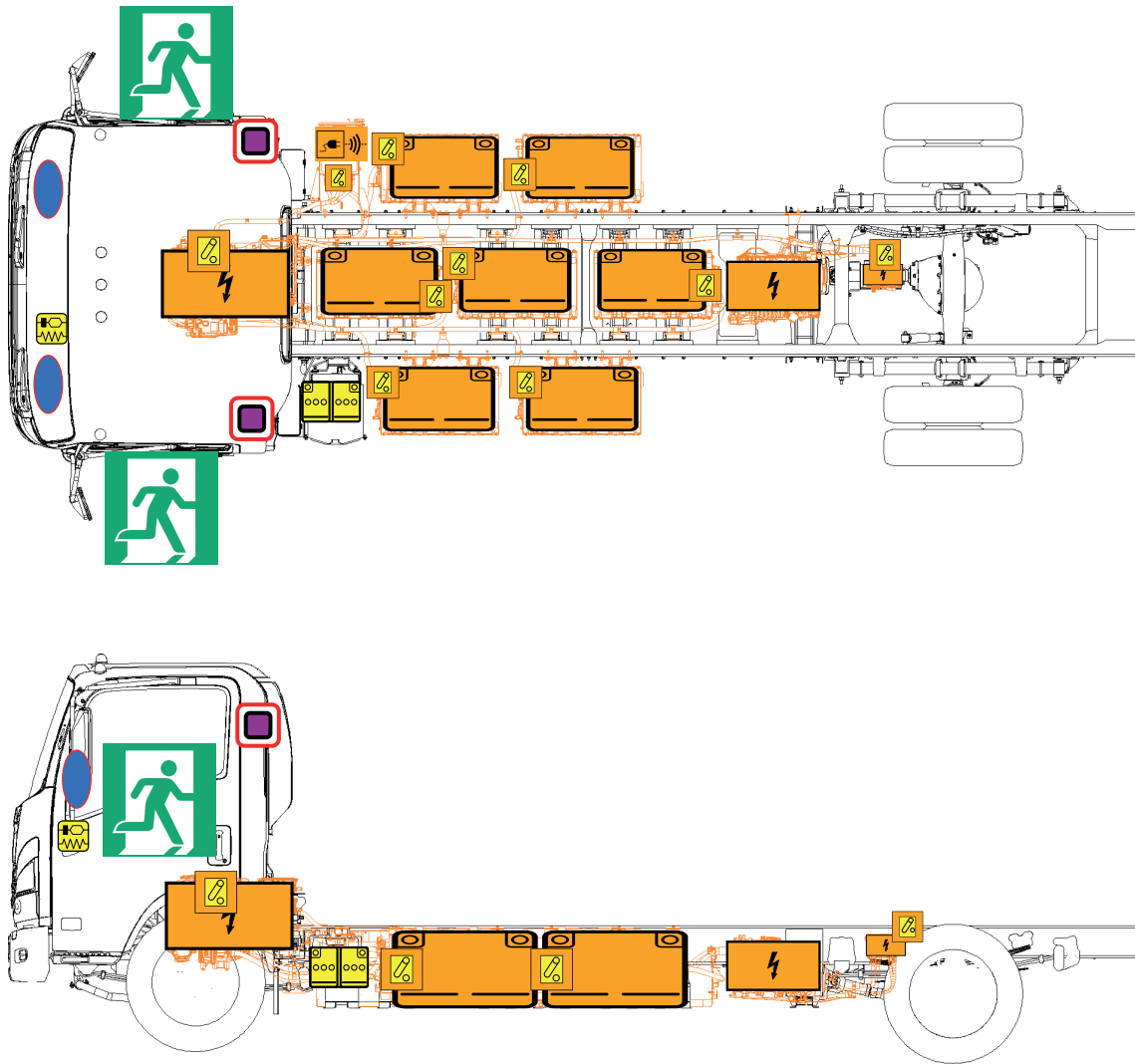
0. Rescue Sheet

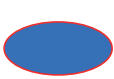


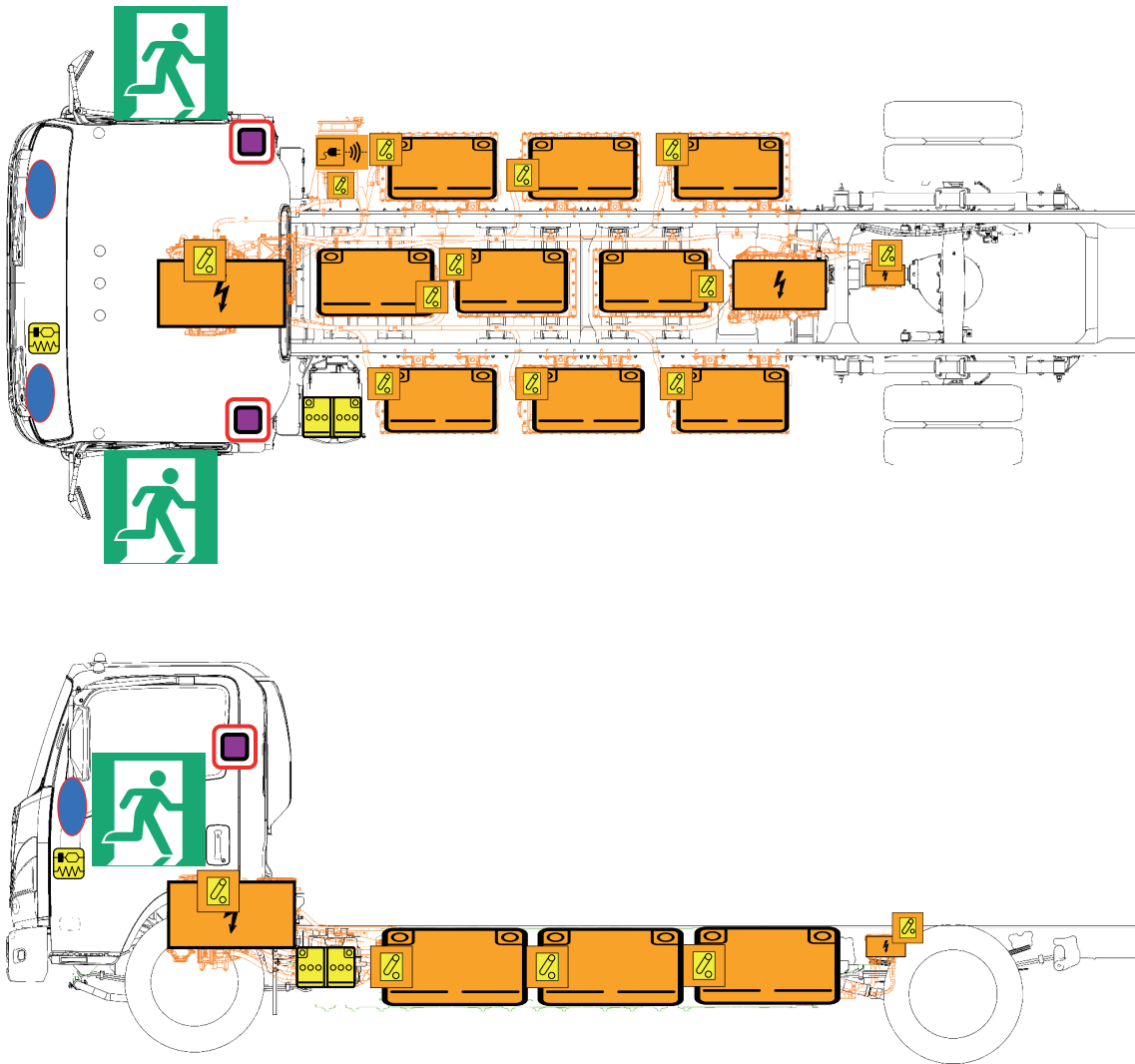
	Charge Port		High-Voltage Li-Ion Battery		High-Voltage Component		Low-Voltage Battery
	High-Voltage Power		Exits		Disconnect High Voltage		Supplemental Restraint System (SRS) Frontal Airbag
	Seat belt pretensioner		SRS control unit				



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1. Identification / recognition



Inform Dispatch and all responders that an electric vehicle is involved.



Lack of engine noise does not mean vehicle is off: vehicle movement capability exists until vehicle is fully shut down. Always wear appropriate PPE.

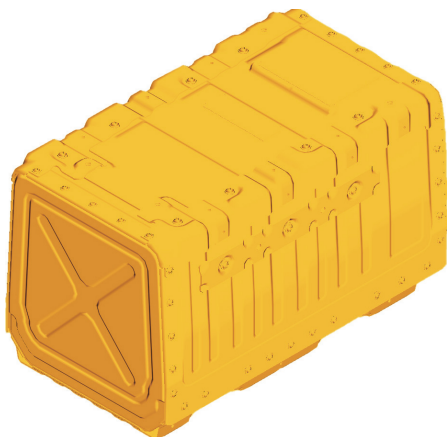
Emblems and Badging



In addition to the exterior image on the cover, the badges on the front panel and left front door can also be used for identification.



High-Voltage Battery Information

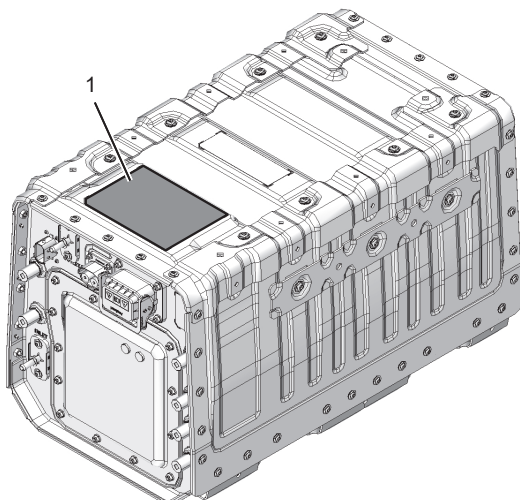


High-voltage batteries are lithium-ion batteries and they are installed between the frames as well as at both sides of the frame.



Battery Warning Label

A battery warning label is located on the front compartment sight shield of each battery.



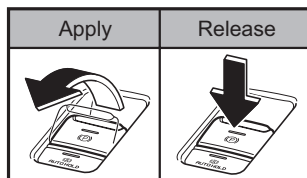
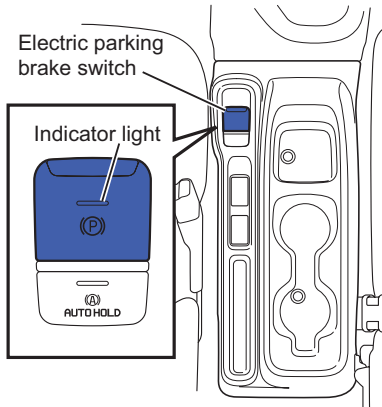
2. Immobilization / stabilization / lifting



IMMOBILIZE VEHICLE

- Block the wheels.
- Follow procedures for conventional vehicles.

Electric Parking Brake (EPB)



When applying the electric parking brake

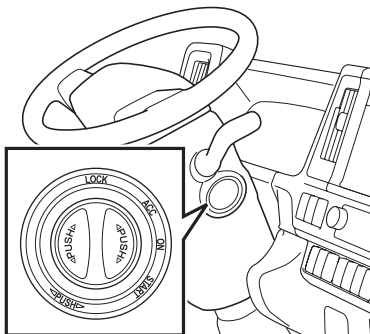
Lifting the electric parking brake switch engages the electric parking brake. The parking brake warning light on the instrument panel and the indicator light on the switch flash until the electric parking brake is engaged. As the parking brake is applied, the warning light and the indicator light will stay on.

When releasing the electric parking brake

Pressing the electric parking brake switch while using your foot to depress the brake pedal releases the electric parking brake. The parking brake warning light on the instrument panel and the indicator light on the switch flash until the electric parking brake is released. After the electric parking brake is released, the warning light and the indicator will turn off.



Shutting Off the EV System



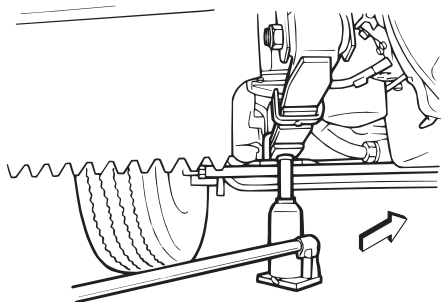
When shutting off the EV system, take the following steps.

1. Bring the vehicle to a complete stop.
2. Apply the electric parking brake.
3. Move the Selector Lever to the "P" (PARK) position and make sure that the shift indicator shows "P".
4. Turn the EV Control Switch to the "LOCK" position.
5. Remove the Keyless entry remote control unit from the cab and at least 30 feet away from the vehicle.

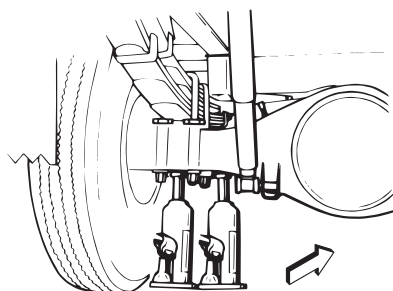


Lifting Points

Front side lifting points



Rear side lifting points



Jack up points of the vehicle

CAUTION

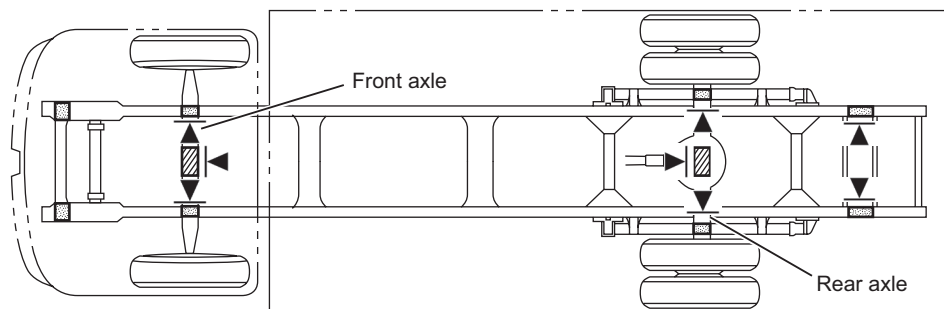
- Securely attach the vehicle jack in the specified position.
- Do not turn the steering wheel while jacking up. There is a risk that the vehicle may come off the jack.

WARNING

- Do not attempt to lift using the high-voltage batteries. Doing so may damage the battery and cause a fire, personal injury or death.

CAUTION

- Securely set the floor jack and rigid rack at the specified position.
- Set the floor jack to the front axle and rear axle.
- Set the rigid rack at the specified position.



 Floor jack point

 Rigid rack supporting position

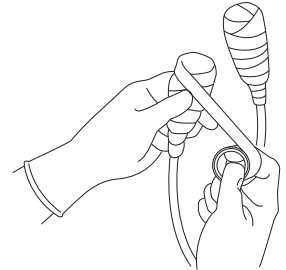
3. Disable direct hazards / safety regulations

EV system shut-down (Decommission) procedure

The EV system high-voltage circuit can be shut-down by removing the MSDs (Manual Service Disconnect). Keep in mind that the location and removal procedure of the MSD differ depending on the vehicle.



- If the abnormal EV system warning light () is ON, high voltage wiring may be in contact with the frame or battery pack. When working on high-voltage parts, be careful not to touch the frame or battery pack, and be sure to use insulated protective equipment to prevent your body from touching them.
- Be sure to wear insulated protective clothing and insulated gloves.
- Never work with moist insulated gloves or clothing moist with sweat, etc.
- Even after the MSDs are removed, the maximum voltage per high voltage battery pack is approximately 400V, which is extremely dangerous if mishandled, so be careful.
- Use a tester to confirm that there is no electric leakage before starting the service work. Do not touch high-voltage terminals that do not have insulation coating without wearing insulated protective equipment.
- If high-voltage connectors and terminals are removed, insulate them by wrapping them with insulating tape.



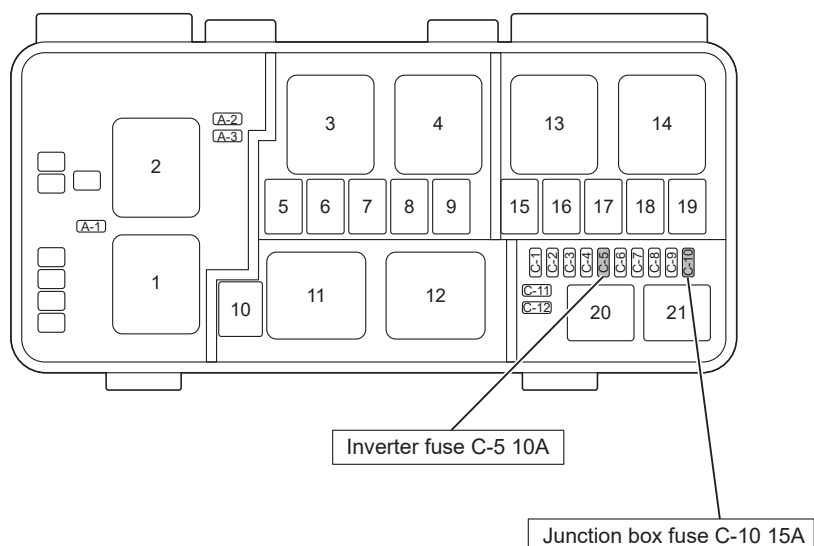
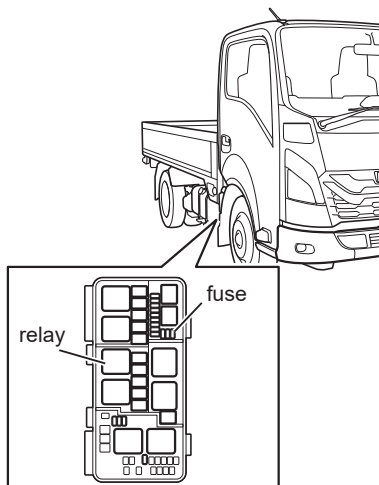
EV system shut-down (Decommission) procedure

Stop the system using one of the following methods.

1-1 Turn OFF the EV system control switch.

- Remove the Keyless Remote control unit from the vehicle and keep it at least 30 feet away from the cab to prevent the EV system from activating.

1-2 Remove the inverter fuse and HV_JB fuse from the fuse and relay box.
(When the service work cannot be done inside the cab)

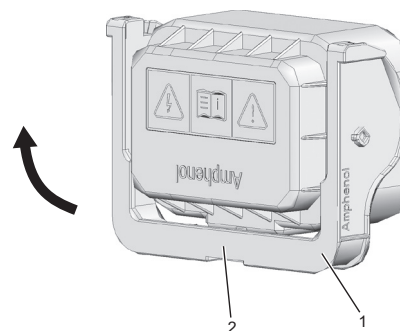
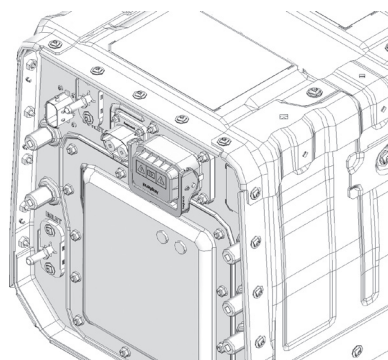


2. Remove or cut the negative terminals of the battery for auxiliary 12V battery.



- It is extremely dangerous to remove the MSDs when the negative terminal of 12V battery for auxiliary equipment cannot be removed. Please take sufficient insulation measures before starting the work.

3. Remove each high-voltage battery pack MSD from the high-voltage battery pack.
- Be sure to wear insulated protective clothing and insulated gloves.
 - Slide the claw (2) and raise the lever (1) until it is horizontal.
 - Remove the same number of MSDs as the number of the high-voltage battery packs installed.



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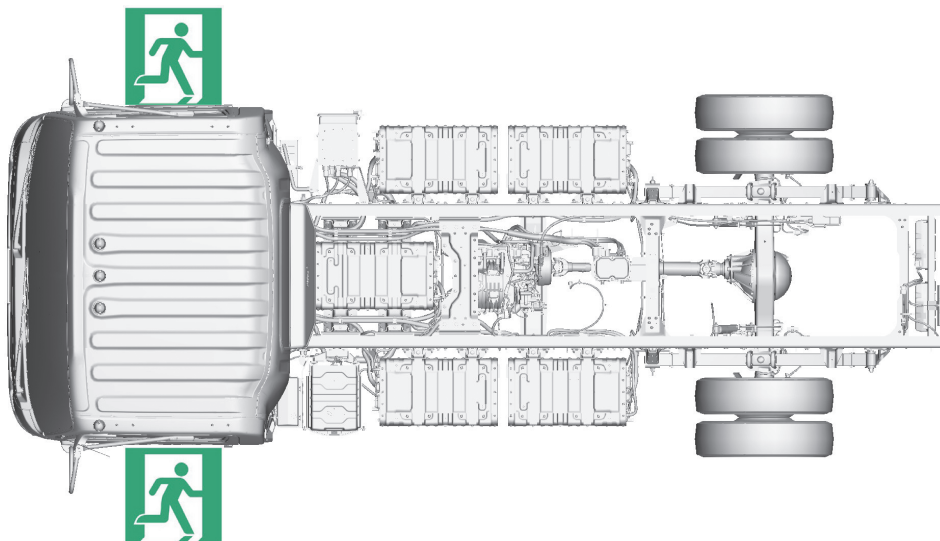


- After removing an MSD, the worker who removed the MSD should manage it to prevent other workers from accidentally connecting the MSD.
- In order to discharge the capacitors in the inverter, wait for 5 minutes or more after removing the MSD, and then start working. If the service work is started without completing discharge, there is the risk of an electric shock.
- Be careful not to accidentally touch the terminals of the MSD connection with your fingers or bare hands.

4. Access to the occupants

How to access inside the vehicle

There are two (2) front doors for access on both sides of the front of the vehicle.



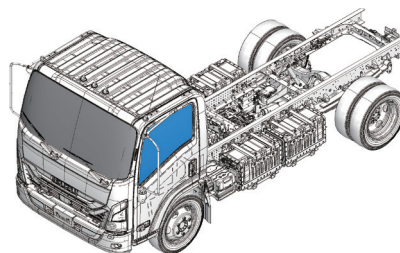
Vehicle Glass



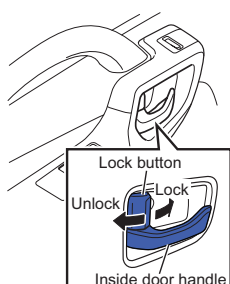
- The windshield is made of Laminated Glass



- The door windows and rear window are made of Tempered Glass



Opening a Locked Side Door

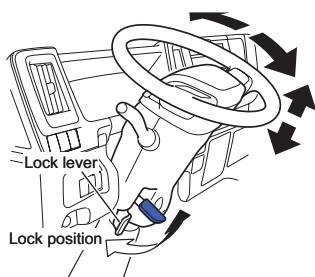


Push the lock button forward to lock the door; pull the lock button backward to unlock it.
When the doors are locked, the inside door handles will not open them.

NOTE

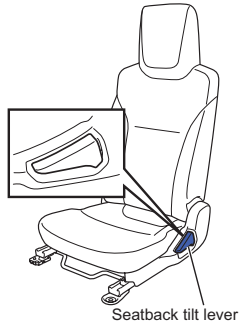
- When unlocked in this way, only the door you are trying to unlock is unlocked.

Steering Column Tilt and Telescoping Control



1. Lift the lock lever toward you to unlock the steering column.
2. Sit in the correct driving position, and then move the steering wheel up and down and forward and backward to select the optimum steering wheel position.
3. Firmly lock the steering wheel at the selected position by moving the lock lever to the lock position.

Seat Controls



Forward/Backward Adjustment

While raising the adjustment lever, move the seat forward or backward. Release the lever when the seat is in the desired position. After making adjustments, try to move the seat back and forth to check that it is fully locked.

Reclining Adjustment

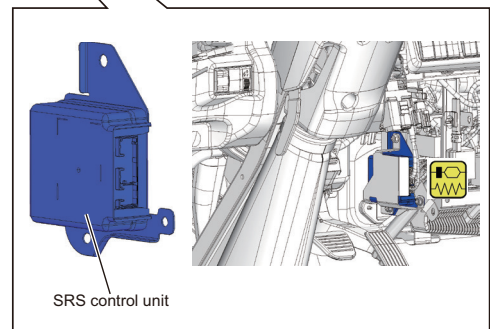
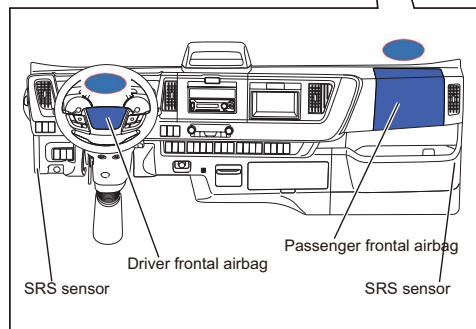
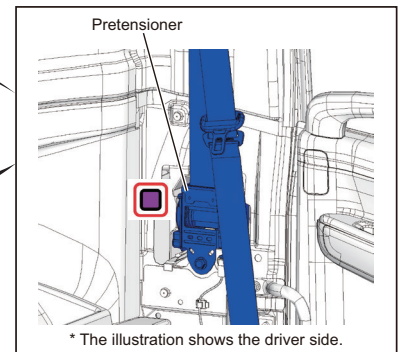
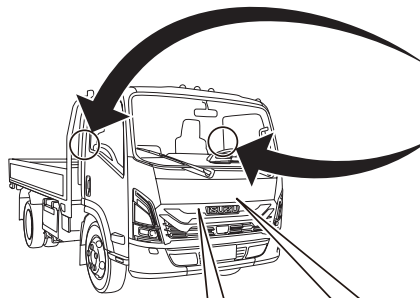
To recline the seatback, raise the seatback tilt lever and gently lean back to the desired position.

To move the seatback forward, lean forward with your back slightly clear of the seatback and raise the lever. After making adjustments, check that the seatback is fully locked.

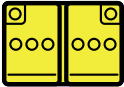
Occupant Restraint Systems

The SRS system mainly consists of the following components, and the wiring color is identified with yellow.

- Driver frontal airbag
- Passenger frontal airbag
- Airbag sensors
- Pretensioners
- SRS control unit



5. Stored energy / liquids / gases / solids

	Low-Voltage Lead Acid Chemistry Battery
	High-Voltage Lithium-Ion Chemistry Battery
	High-Voltage Warning, potential for electric shock
	Gases emitted from the battery pack are flammable
	Gases emitted from the battery pack are toxic
	Skin contact may cause irritation. Prolonged contact with electrolyte mixture may result in more severe irritation. Flush contaminated skin with plenty of water.
	Coolant leaking inside the battery pack can become unstable and possibly cause a fire.

6. In case of fire

	High Voltage Warning, potential for electric shock
	If battery cells reach high enough temperature, they can vent and release electrolyte. Battery electrolyte is flammable.
	Gases emitted from the battery pack are toxic
	Skin contact may cause irritation. Prolonged contact with electrolyte mixture may result in more severe irritation. Flush contaminated skin with plenty of water.
	Potential for eye, nose, and throat irritation with prolonged exposure.

	When spraying water, use a large amount of water from a fire hydrant, etc., as extinguishing a fire with a small amount of water can be dangerous. Maintain the safe distance and sufficiently take care not to get an electric shock when applying water.
	Use an ABC or class D fire extinguisher (effective for general, oil and grease, and electrical fires) to extinguish the fire.
 	Potential for Battery Re-Ignition.

7. In case of submersion

How to deal with submerged vehicles

If the electric vehicle is submerged in water, follow the procedures below. When touching the vehicle, be sure to wear the appropriate personal protective equipment (PPE).

Do not attempt to shut-down (decommission) a submerged vehicle. This could cause personal injury or death.

- Remove the vehicle out of the water, avoiding contact with the high voltage system and batteries.
- Allow water to drain from the vehicle. If it is difficult to drain, vent the passenger compartment and leave the vehicle for a long period of time until it is completely dry.

When draining water from the vehicle, do not touch the vehicle without the proper PPE as there is a risk of electric shock.

- After the vehicle is dry and wearing the appropriate PPE, follow the EV system shut-down (Decommission) procedure described in this manual.

Do not store a recovered vehicle indoors. Damaged high-voltage batteries can produce flammable gas and fire which can lead to serious injury or death.

8. Towing / transportation / storage

How to transport a vehicle

If possible, request a professional towing agency to transport the broken vehicle. If it is unavoidable, follow the steps below.

WARNING

- Be sure to apply chock blocks when removing the rear axle shaft. The vehicle may start moving, and that may cause a serious accident.
- When towing, lift the front wheels or 4 wheels. The drive motor may generate electricity, and that may cause a fire.

ADVICE

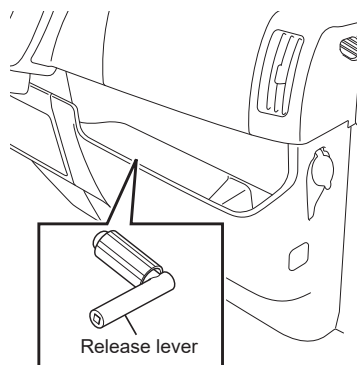
- When towing with the front wheels lifted, be sure to pull out the rear axle shaft. If the vehicle is towed without the drive shaft disconnected, the EV system components may be damaged.
- When towing with the 4 wheels lifted, apply the parking brake and securely fix the vehicle to prevent it from shifting/moving during transportation.

Release of electric parking brake (When EV Control Switch cannot be turned ON)

WARNING

- Do not manually release the electric parking brake except when the vehicle is towed by a specialized tow truck or when temporarily moving the vehicle.
- Do not release the electric parking brake manually while on a slope. The vehicle may roll uncontrolled and result in an accident, serious injury or death.
- After manually releasing it, move the vehicle to a safe location and immediately contact the nearest Isuzu Dealer.

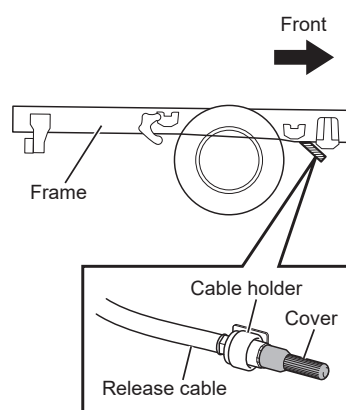
1. Put the Selector Lever in "P" position and securely chock the wheels.
2. Take out the electric parking brake release lever stored in the passenger side storage pocket.



3. Remove the cover from the tip of the release cable.

NOTE

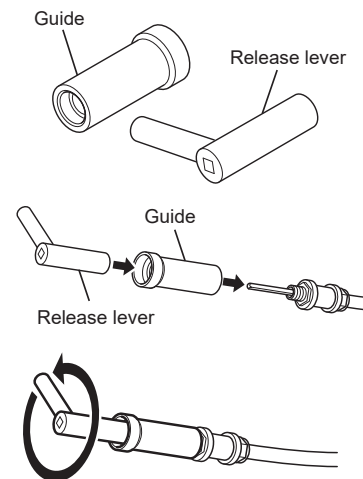
- The release cable is located on the side of the right-rear frame.



4. Remove the guide from the electric parking brake release lever.
5. Attach the guide and then the release lever to the release cable, in this order.
6. Press in on the release lever as you turn it to the left 35 times.

NOTE

- If the electric parking brake is not released, then turn the release lever an additional 5 turns at a time to the left until the electric parking brake is released.



7. After the electric parking brake is released, attach the cover to the tip of the release cable and return the release cable to the cable holder.

Release of electric parking brake (When EV Control Switch can be turned ON)

1. Turn the EV Control Switch OFF while pressing the electric parking brake switch.

Front End Towing

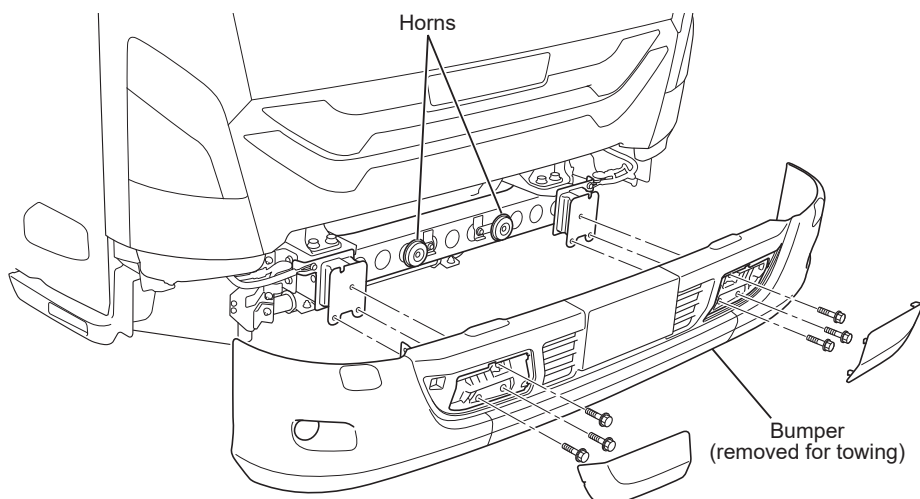
Front Wheels Off Ground

To prepare a disabled vehicle for front end towing with front wheels raised off the ground, the following steps are necessary:

- Block the rear wheels of the disabled vehicle.
- Disconnect the driveshaft at the rear axle. Secure the driveshaft to the frame or crossmember.

ADVICE

- When towing, disconnect the driveshaft at the rear axle to ensure the EV system components are not damaged.
1. If there is damage or suspected damage to the rear axle, remove the axle shafts. Cover the hub openings to prevent the loss of lubricant or entry of dirt or foreign objects.
 2. Place a 4 in (10 cm) wood beam against the towing guide behind the bumper. (If no 4 in (10 cm) wood beam is available, then remove the bumper.) Ensure towing chains do not come into contact with the horns or the bumper. (If towing chains contact the bumper, then remove the bumper.)



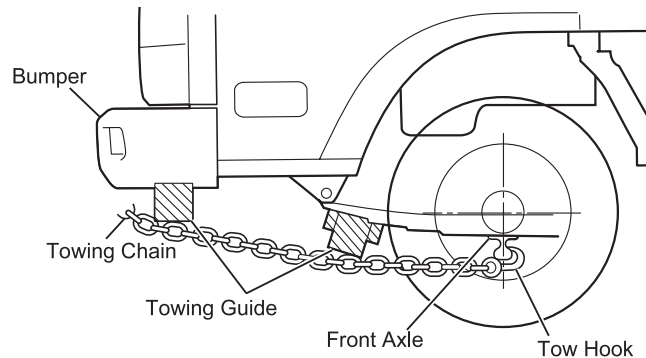
After Towing

After towing the vehicle, block the rear wheels and install axle shafts and/or driveshaft. Apply the electric parking brake before disconnecting from the towing vehicle.

All Wheels On the Ground

Your vehicle may be towed on all wheels provided the steering is operable. Remember that power steering and brakes will not have power assist. There must be a tow bar installed between the tow vehicle and the disabled vehicle.

Towing with all wheels on the ground



To prepare a disabled vehicle for front end towing with all wheels on the ground, the following steps are necessary:

- Block the wheels of the disabled vehicle.
- Disconnect the driveshaft at the rear axle. Secure the driveshaft to the frame or crossmember.

ADVICE

- When towing, disconnect the driveshaft at the rear axle to ensure the EV system components are not damaged.

Provide wood blocking to prevent towing chains and bar from coming into contact with the bumper.

If there is damage or suspected damage to the rear axle, remove the axle shafts.

Cover the hub openings to prevent the loss of lubricant or entry of dirt or foreign objects.

After Towing

After towing the vehicle, block the rear wheels and install axle shafts or driveshaft.

Apply the electric parking brake before disconnecting from the towing vehicle. Check and fill the rear axle with oil, if required.

Rear End Towing

When towing a vehicle with rear wheels raised, secure the steering wheel to maintain straight-ahead position. Make certain that the front axle is not loaded beyond the front axle gross axle weight rating (GAWR) as indicated on the vehicle's "VIN, Weight Rating, and Greenhouse Gas (GHG) Emissions Plate".

Special Towing Instructions

- All state and local laws regarding such items as warning signals, night illumination, speed, etc., must be followed.
- Safety chains must be used.
- No vehicle should ever be towed over 55 MPH (88 km/h).
- Loose or protruding parts of damaged vehicles should be secured prior to moving.
- A safety chain system completely independent of the primary lifting and towing attachment must be used.
- Operators should refrain from going under a vehicle which is being lifted by the towing equipment unless the vehicle is adequately supported by safety stands.
- No towing operation which for any reason jeopardizes the safety of the wrecker operator or any bystanders or other motorists should be attempted.

Post-Crash Vehicle Storage

WARNING

If a high-voltage battery is damaged or suspected of being damaged (e.g., involved in a collision), do not touch any of the vehicle's high-voltage components.

Continuing to drive the vehicle could cause further damage to the EV system (e.g., water can enter the battery), increasing the risk of serious injury or death. Have the vehicle inspected as soon as possible by the nearest Isuzu Dealer.

9. Explanation of pictograms used

	Electric Vehicle		General warning sign		Warning, High Voltage
	High-voltage Battery		Lifting Points		Thermal Imaging Camera
	Flammable		Toxic		Corrosive
	Injury Risk		Use Water		Low-Voltage Battery
	Disconnect High Voltage		Use ABC powder		SRS Frontal Airbag
	Seat belt pretensioner		SRS control unit		High-Voltage Component

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